

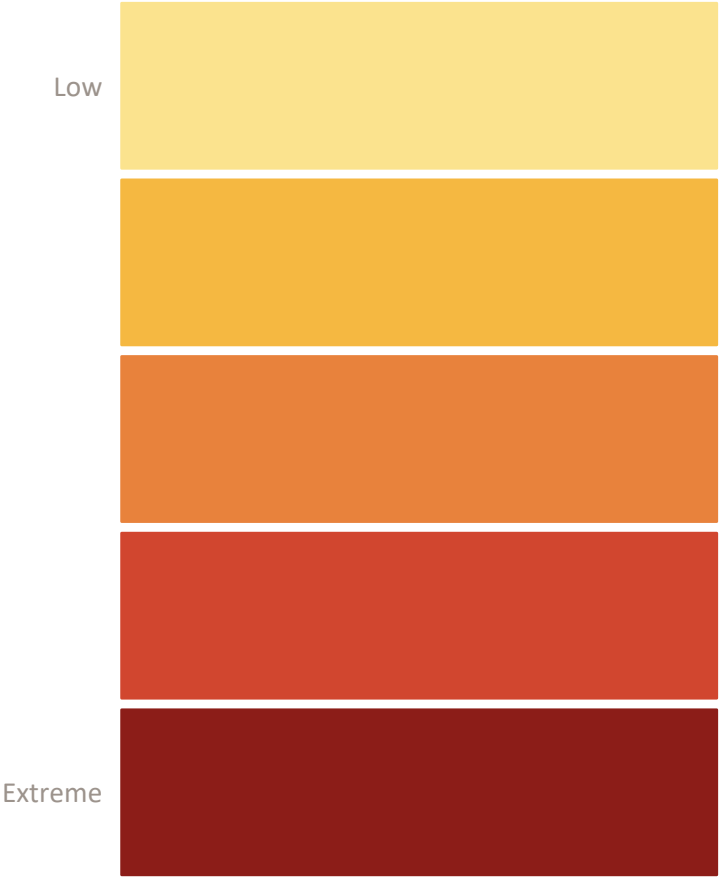
Theme: AI for Climate Change

HeatShield

From heatwave alerts to hyperlocal action.

Team members

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Outline

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Problem Statement / Idea

Heat risk is hyperlocal; response is citywide

4

Objectives

Map, forecast, place relief, advise

5

Proposed Solution / Architecture

The pipeline and the stack

6

Unique Features / Innovation

The action layer, not the map

7

Expected Outcomes

Deliverables and impact

8

Conclusion

Why this, why now

The gap we are closing

**“Heat mapping in India is
a solved problem.
Heat action isn’t.”**

Sections 3 to 8 follow one line of argument: the risk is hyperlocal, the response is not, and the missing piece is the step from a map to a decision.

3 Problem Statement / Idea

The problem

Indian heatwave response is citywide and blind, when heat risk is hyperlocal and predictable. A city issues one red-alert threshold for the whole municipality — but inside that city, risk varies block to block. A dense, tin-roofed, tree-poor pocket can run several degrees hotter than a leafy planned area a few kilometres away, and it is usually home to the people least able to cope: outdoor workers, the elderly, those without cooling.

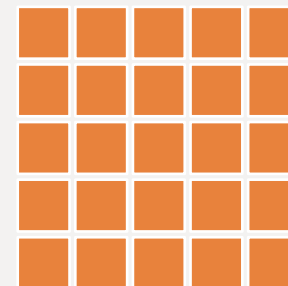
Why it matters

Heat is one of India's fastest-growing climate killers. Ahmedabad's 2013 Heat Action Plan — South Asia's first — is credited with averting roughly 1,100+ deaths a year, and worked well enough that the NDMA told 23 heat-prone states to build their own. But even good Heat Action Plans mostly stop at citywide alerts.

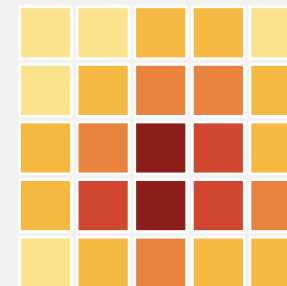
Current gaps in existing solutions

Hyperlocal heat mapping already works — SEEDS with Microsoft, IIT Mandi's building-level maps — which proves the mapping problem is solvable. But most of that work stops at the map. It rarely closes the loop into “place the cooling centre here, today.”

One alert, whole city



What is actually happening



vs

Same city, same hour. The alert is uniform; the risk is not — and the hottest pockets hold the people least able to cope.

Our idea, in one line

HeatShield turns a citywide heatwave alert into a ward-level action plan — mapping who is at risk, forecasting it forward, and recommending exactly where to send relief.

4

Objectives

Key goals — one chain, four links

1

Map

Heat vulnerability at grid and ward level — not one number for the whole city.

2

Forecast

Push that risk 24–72 hours forward on live weather, so it updates daily instead of sitting frozen.

3

Place

Recommend where cooling relief should go, so the resources a city already has protect the most people.

4

Advise

Generate ward-level advisories a municipal officer can act on directly.

Expected impact

Better-targeted relief without needing more resources — the same number of cooling centres and water points protecting more people, because they sit where risk actually concentrates. And a faster, more specific municipal response than a citywide alert allows.

End beneficiaries

- Municipal disaster management cells and city health departments — the direct users
- Urban poor, outdoor workers and elderly residents — the people hardest to reach through generic alerts

5

Proposed Solution / Architecture

Data ingestion

Static vulnerability layer

Satellite land-surface temperature, tree cover, building density, census vulnerability. Precomputed once per city.

Live weather forecast

Open-Meteo 24–72 h temperature and humidity, refreshed daily.

Risk engine

Per-cell risk score

Fuses the static layer with today's forecast into a grid-level heat-risk surface that updates daily — not a frozen historical atlas.

Action layer

Interactive map

Renders the risk surface; drill into any cell or ward for its vulnerability breakdown.

Siting optimizer

Reads the surface against existing hospitals and water points, then recommends where new relief should go.

Advisory generator

An LLM turns a ward's risk data into a short, plain-language advisory an officer can send.

Key technologies

Next.js + TypeScript frontend · MapLibre GL for the interactive map · Neon (Postgres) for storage · hand-written coverage-optimisation algorithm · Gemini API for advisory text · deployed on Vercel.

Scalability and feasibility

The architecture is city-agnostic. The same pipeline runs for any Indian city once its static vulnerability layer is precomputed — built to extend city by city, not rebuilt per city.

Unique Features / Innovation

The differentiator is the action layer, not the map.

Hyperlocal heat mapping already exists in India — SEEDS with Microsoft, IIT Mandi. HeatShield's contribution is closing the loop from risk map to resource placement: a siting optimizer most comparable projects do not attempt.



Live, not static

The risk surface moves with the daily forecast instead of being a fixed historical vulnerability atlas. It answers “where is it dangerous today,” not just “where is it historically hot.”



Built for the last mile

Auto-generated ward-level advisories mean the output is something a municipal officer can act on directly — not one more dashboard for analysts.



Optimisation, not just visualisation

A coverage-maximising algorithm turns the map into a ranked list of placements, with the population gain each one buys made explicit.

Expected Outcomes

Deliverables

- A working web dashboard with a live, interactive ward-level heat-risk map
- A siting-recommendation tool showing coverage gained against current facility placement
- Auto-generated advisory text for any selected ward

Short-term impact

A demonstrable prototype for one Indian city showing measurably better relief placement than the status quo — a concrete “current placement protects X people, recommended placement protects Y” comparison, computed live.

Long-term impact

A template municipalities could adopt city by city, to make Heat Action Plans hyperlocal and actionable rather than citywide and generic.

The number the demo produces

Coverage of the at-risk population, before and after the optimizer’s recommended placements:

Current placement



With recommended placements



Illustrative — the live figure is computed per city and per day, not fixed.

Scope to scale

- More cities, by precomputing their vulnerability layers
- Beyond heat — flooding, air quality — on the same map → forecast → optimise → advise pattern

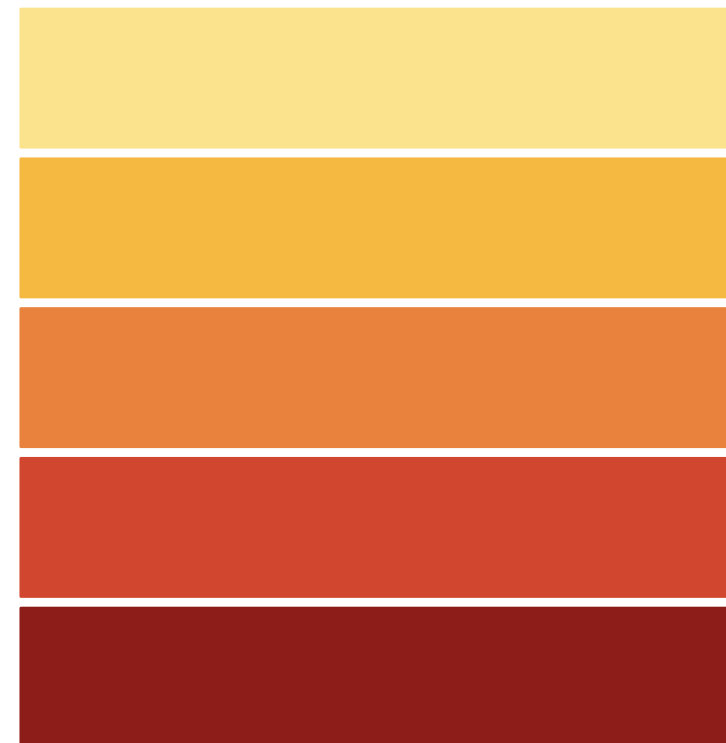
Heat mapping in India is a solved problem.

Heat action isn't.

HeatShield takes proven hyperlocal risk-mapping and closes the gap the existing tools leave open — turning a citywide alert into a live, ward-level plan for exactly where to send relief.

Why it should be selected

- Not novel for novelty's sake — it builds on India's own Ahmedabad Heat Action Plan legacy
- It targets a documented life-and-death gap, not a hypothetical one
- Scoped as a working, demoable system — not a single model or a static dashboard



HeatShield

From heatwave alerts to hyperlocal action.